

Polaron Transformation

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1 Model Hamiltonian and Redfield Equation

In this section we summarize some definitions and equations that will be used for subsequent derivations, including the model Hamiltonian, bath correlation function, spectral density, and the Redfield equation. The derivations for those expressions can be found in previous notes. Note that we use a different definition of spectral density and Redfield tensor here, which is more commonly used in recent research papers.

The model Hamiltonian is a multi-level system linearly coupled to a harmonic oscillator bath,

$$\begin{aligned}
 \hat{H} &= \hat{H}_S + \hat{H}_B + \hat{H}_{SB} \\
 &= \sum_m (\varepsilon_m + \lambda_m) |m\rangle\langle m| + \sum_{m \neq n} V_{mn} |m\rangle\langle n| \\
 &\quad + \sum_j \hbar \omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right) \\
 &\quad - \sum_m \sum_j \gamma_{j,m} (\hat{b}_j^\dagger + \hat{b}_j) |m\rangle\langle m|
 \end{aligned} \tag{1}$$

where ε_m is the energy of state $|m\rangle$ in an uncoupled system, λ_m is the bath reorganization energy, V_{mn} is the coupling between states $|m\rangle$ and $|n\rangle$, $\gamma_{j,m}$ is the coupling strength between state $|m\rangle$ and mode j .

The system-bath coupling can be factorized as $\hat{H}_{SB} = \sum_m \hat{S}_m \otimes \hat{B}_m$, where $\hat{S}_m = |m\rangle\langle m|$ is an operator only act on the system, and $\hat{B}_m = -\sum_j \gamma_{j,m} (\hat{b}_j^\dagger + \hat{b}_j)$ is an operator only act on the bath. The bath correlation function is defined as

$$C_{mn}(t) = \text{Tr}_B \left(\hat{B}_m(t) \hat{B}_n(0) \hat{\rho}_{B,\text{eq}} \right) \tag{2}$$

where $\hat{B}_m(t) = e^{\frac{i}{\hbar} \hat{H}_B t} \hat{B}_m e^{-\frac{i}{\hbar} \hat{H}_B t}$, and $\hat{\rho}_{B,\text{eq}} = \frac{e^{-\beta \hat{H}_B}}{\text{Tr}_B e^{-\beta \hat{H}_B}}$.

For harmonic oscillator bath, $C_{mn}(t)$ can be calculated analytically as

$$C_{mn}(t) = \frac{\hbar}{\pi} \int_0^\infty d\omega J_{mn}(\omega) \left[\cos(\omega t) \coth \frac{\beta \hbar \omega}{2} - i \sin(\omega t) \right] \tag{3}$$

where

$$J_{mn}(\omega) = \frac{\pi}{\hbar} \sum_\alpha \gamma_{j,m} \gamma_{j,n} \delta(\omega - \omega_\alpha) \tag{4}$$

is the spectral density. The reorganization energy can also be expressed using the spectral density as

$$\lambda_m = \frac{1}{\pi} \int_0^\infty \frac{J_{mm}(\omega)}{\omega} d\omega \quad (5)$$

Let $|a\rangle, |b\rangle, \dots$ represent the eigenstate of the system Hamiltonian $\hat{H}_S$, and $E_a, E_b, \dots$ be the energies of the eigenstates. The matrix elements of the $\hat{S}_m$ operators are

$$S_{ab}^{(m)} = \langle a | \hat{S}_m | b \rangle = \langle a | m \rangle \langle m | b \rangle \quad (6)$$

Let $G_{mn}(\omega)$ be the half Fourier transformation of $C_{mn}(t)$,

$$G_{mn}(\omega) = \int_0^\infty dt e^{i\omega t} C_{mn}(t) \quad (7)$$

The Redfield tensor in the eigenstate representation is given by

$$\mathcal{R}_{abcd} = -\delta_{bd} \sum_e \Gamma_{aeec} - \delta_{ac} \sum_e \Gamma_{beed}^* + \Gamma_{dbac} + \Gamma_{cabd}^* \quad (8)$$

where

$$\Gamma_{abcd} = \sum_{m,n} S_{ab}^{(m)} S_{cd}^{(n)} G_{mn}(\omega_{dc}) \quad (9)$$

and $\omega_{ab} = (E_a - E_b)/\hbar$. Make the substitution, we have

$$\begin{aligned} \mathcal{R}_{abcd} = & -\delta_{bd} \sum_{m,n} \sum_e S_{ae}^{(m)} S_{ec}^{(n)} G_{mn}(\omega_{ce}) - \delta_{ac} \sum_{m,n} \sum_e S_{eb}^{(m)} S_{de}^{(n)} G_{mn}^*(\omega_{de}) \\ & + \sum_{m,n} S_{db}^{(m)} S_{ac}^{(n)} G_{mn}(\omega_{ca}) + \sum_{m,n} S_{ac}^{(m)} S_{db}^{(n)} G_{mn}^*(\omega_{db}) \end{aligned} \quad (10)$$

and the Redfield equation is

$$\frac{d}{dt} \rho_{ab} = -i\omega_{ab} \rho_{ab} + \frac{1}{\hbar^2} \sum_{c,d} \mathcal{R}_{abcd} \rho_{cd} \quad (11)$$

2 Polaron Transformation of the Hamiltonian

The Redfield equation is derived based on perturbative expansion of the system-bath coupling to the second order. Therefore, it is only valid in the weak system-bath coupling limit, where

$$\lambda_m \ll \omega_{ab} \quad \text{or} \quad \lambda_m \ll V_{mn} \quad (12)$$

The Redfield equation becomes unreliable when this condition is not met. One way to resolve this issue is to apply a canonical transformation called **polaron transformation**, or **Lang-Firsov transformation**, which describes the system in the polaron frame such that the system is dressed by the bath. The dressed system takes the major effects of the system-bath coupling into consideration, whereas the reduced system-bath coupling is weakened to the regime that second-order perturbation is applicable.

The polaron transformation of the total Hamiltonian is

$$\hat{\tilde{H}} = e^{\hat{G}} \hat{H} e^{-\hat{G}} \quad (13)$$

where

$$\hat{G} = - \sum_m \sum_j \frac{\gamma_{j,m}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) |m\rangle \langle m| \quad (14)$$

is the generating operator of the transformation. It is clear that this transformation is unitary because $\hat{G}^\dagger = -\hat{G}$.

Now we derive how operators $\hat{b}_j$, $\hat{b}_j^\dagger$, and $|m\rangle \langle n|$ are transformed under this transformation. The derivation is based on the Baker-Campbell-Hausdorff formula,

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \sum_{l=0}^{\infty} \frac{1}{l!} [\hat{A}, \hat{B}]_l = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2} [\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{6} [\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots \quad (15)$$

Recall that the Boson commutation relation is

$$[\hat{b}_j, \hat{b}_k^\dagger] = \delta_{jk}, \quad [\hat{b}_j, \hat{b}_k] = [\hat{b}_j^\dagger, \hat{b}_k^\dagger] = 0 \quad (16)$$

The commutator of $\hat{G}$ with $\hat{b}_j$, $\hat{b}_j^\dagger$, and $|m\rangle \langle n|$ are

$$\begin{aligned} [\hat{G}, \hat{b}_j] &= \left[- \sum_m \sum_k \frac{\gamma_{k,m}}{\hbar\omega_k} (\hat{b}_k^\dagger - \hat{b}_k) |m\rangle \langle m|, \hat{b}_j \right] \\ &= - \sum_m \sum_k \frac{\gamma_{k,m}}{\hbar\omega_k} ([\hat{b}_k^\dagger, \hat{b}_j] - [\hat{b}_k, \hat{b}_j]) |m\rangle \langle m| \\ &= - \sum_m \sum_k \frac{\gamma_{k,m}}{\hbar\omega_k} (-\delta_{jk}) |m\rangle \langle m| \\ &= \sum_m \frac{\gamma_{j,m}}{\hbar\omega_j} |m\rangle \langle m| \end{aligned} \quad (17)$$

Similarly,

$$[\hat{G}, \hat{b}_j^\dagger] = \sum_m \frac{\gamma_{j,m}}{\hbar\omega_j} |m\rangle \langle m| \quad (18)$$

$$\begin{aligned} [\hat{G}, |m\rangle \langle n|] &= \left[- \sum_{m'} \sum_j \frac{\gamma_{j,m'}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) |m'\rangle \langle m'|, |m\rangle \langle n| \right] \\ &= - \sum_{m'} \sum_j \frac{\gamma_{j,m'}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) (|m'\rangle \langle m'| m\rangle \langle n| - |m\rangle \langle n| m'\rangle \langle m'|) \\ &= - \sum_{m'} \sum_j \frac{\gamma_{j,m'}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) (\delta_{mm'} |m'\rangle \langle n| - \delta_{m'n} |m\rangle \langle m'|) \\ &= - \sum_j \frac{\gamma_{j,m}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) |m\rangle \langle n| + \sum_j \frac{\gamma_{j,n}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) |m\rangle \langle n| \\ &= - \sum_j \frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) |m\rangle \langle n| \end{aligned} \quad (19)$$

Specifically, $[\hat{G}, |m\rangle\langle m|] = 0$.

The next nested commutators are

$$[\hat{G}, [\hat{G}, \hat{b}_j]] = [\hat{G}, [\hat{G}, \hat{b}_j^\dagger]] = \sum_m \frac{\gamma_{j,m}}{\hbar\omega_j} [\hat{G}, |m\rangle\langle m|] = 0 \quad (20)$$

$$\begin{aligned} [\hat{G}, [\hat{G}, |m\rangle\langle n|]] &= - \sum_j \frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) [\hat{G}, |m\rangle\langle n|] \\ &= \sum_j \frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) \sum_k \frac{\gamma_{k,m} - \gamma_{k,n}}{\hbar\omega_k} (\hat{b}_k^\dagger - \hat{b}_k) |m\rangle\langle n| \\ &= \left[- \sum_j \frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) \right]^2 |m\rangle\langle n| \end{aligned} \quad (21)$$

In general,

$$[\hat{G}, |m\rangle\langle n|]_l = \left[- \sum_j \frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) \right]^l |m\rangle\langle n| \quad (22)$$

The transformation of the operators $\hat{b}_j$, $\hat{b}_j^\dagger$, and $|m\rangle\langle n|$ are thus

$$e^{\hat{G}} \hat{b}_j e^{-\hat{G}} = \hat{b}_j + \sum_m \frac{\gamma_{j,m}}{\hbar\omega_j} |m\rangle\langle m| \quad (23)$$

$$e^{\hat{G}} \hat{b}_j^\dagger e^{-\hat{G}} = \hat{b}_j^\dagger + \sum_m \frac{\gamma_{j,m}}{\hbar\omega_j} |m\rangle\langle m| \quad (24)$$

$$\begin{aligned} e^{\hat{G}} |m\rangle\langle n| e^{-\hat{G}} &= \sum_{l=0}^{\infty} \frac{1}{l!} \left[- \sum_j \frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) \right]^l |m\rangle\langle n| \\ &= \exp \left[- \sum_j \frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) \right] |m\rangle\langle n| \\ &= \exp \left[- \sum_j \frac{\gamma_{j,m}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) \right] \exp \left[\sum_j \frac{\gamma_{j,n}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) \right] |m\rangle\langle n| \end{aligned} \quad (25)$$

Defining a unitary operator

$$\hat{\theta}_m = \exp \left[\sum_j \frac{\gamma_{j,m}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j) \right] \quad (26)$$

we have a compact expression

$$e^{\hat{G}} |m\rangle\langle n| e^{-\hat{G}} = \hat{\theta}_m^\dagger \hat{\theta}_n |m\rangle\langle n| \quad (27)$$

With these expressions, we can easily derive the transformation of the Hamiltonian. For the system Hamiltonian, $\hat{H}_S$, clearly

$$e^{\hat{G}} \hat{H}_S e^{-\hat{G}} = \sum_m (\varepsilon_m + \lambda_m) |m\rangle\langle m| + \sum_{m \neq n} V_{mn} \hat{\theta}_m^\dagger \hat{\theta}_n |m\rangle\langle n| \quad (28)$$

For the bath Hamiltonian $\hat{H}_B$,

$$\begin{aligned}
e^{\hat{G}} \hat{H}_B e^{-\hat{G}} &= \sum_j \hbar \omega_j \left(e^{\hat{G}} \hat{b}_j^\dagger e^{-\hat{G}} e^{\hat{G}} \hat{b}_j e^{-\hat{G}} + \frac{1}{2} \right) \\
&= \sum_j \hbar \omega_j \left[\left(\hat{b}_j^\dagger + \sum_m \frac{\gamma_{j,m}}{\hbar \omega_j} |m\rangle \langle m| \right) \left(\hat{b}_j + \sum_n \frac{\gamma_{j,n}}{\hbar \omega_j} |n\rangle \langle n| \right) + \frac{1}{2} \right] \\
&= \sum_j \hbar \omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right) + \sum_j \sum_m \gamma_{j,m} (\hat{b}_j^\dagger + \hat{b}_j) |m\rangle \langle m| + \sum_j \sum_m \frac{\gamma_{j,m}^2}{\hbar \omega_j} |m\rangle \langle m| \\
&= \sum_j \hbar \omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right) + \sum_m \sum_j \gamma_{j,m} (\hat{b}_j^\dagger + \hat{b}_j) |m\rangle \langle m| + \sum_m \lambda_m |m\rangle \langle m|
\end{aligned} \tag{29}$$

In the last equality, We have used that $\lambda_m = \sum_j \frac{\gamma_{j,m}^2}{\hbar \omega_j}$. Finally, for the system-bath coupling $\hat{H}_{SB}$,

$$\begin{aligned}
e^{\hat{G}} \hat{H}_{SB} e^{-\hat{G}} &= - \sum_m \sum_j \gamma_{j,m} (e^{\hat{G}} \hat{b}_j^\dagger e^{-\hat{G}} + e^{\hat{G}} \hat{b}_j e^{-\hat{G}}) e^{\hat{G}} |m\rangle \langle m| e^{-\hat{G}} \\
&= - \sum_m \sum_j \gamma_{j,m} \left(\hat{b}_j^\dagger + \hat{b}_j + 2 \sum_n \frac{\gamma_{j,n}}{\hbar \omega_j} |n\rangle \langle n| \right) |m\rangle \langle m| \\
&= - \sum_m \sum_j \gamma_{j,m} (\hat{b}_j^\dagger + \hat{b}_j) |m\rangle \langle m| - 2 \sum_m \sum_j \frac{\gamma_{j,m}^2}{\hbar \omega_j} |m\rangle \langle m| \\
&= - \sum_m \sum_j \gamma_{j,m} (\hat{b}_j^\dagger + \hat{b}_j) |m\rangle \langle m| - 2 \sum_m \lambda_m |m\rangle \langle m|
\end{aligned} \tag{30}$$

The final expression of polaron transformed Hamiltonian is

$$\boxed{\hat{\tilde{H}} = e^{\hat{G}} \hat{H} e^{-\hat{G}} = \sum_m \varepsilon_m |m\rangle \langle m| + \sum_{m \neq n} V_{mn} \hat{\theta}_m^\dagger \hat{\theta}_n |m\rangle \langle n| + \sum_j \hbar \omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right)} \tag{31}$$

After the polaron transformation, the bath part of the Hamiltonian remains unchanged, the renormalization and the diagonal system-bath coupling terms disappear from the total Hamiltonian, whereas the coupling V_{mn} becomes a bath-dressed coupling $V_{mn} \hat{\theta}_m^\dagger \hat{\theta}_n$.

Next, we re-partition the transformed Hamiltonian in to the system, bath, and system-bath coupling terms, so that the thermal average of the system-bath coupling is zero. The thermal average of $\hat{\theta}_m^\dagger \hat{\theta}_n$ is

$$\begin{aligned}
\kappa_{mn} &\equiv \text{Tr}_B \left(\hat{\rho}_{B,\text{eq}} \hat{\theta}_m^\dagger \hat{\theta}_n \right) \\
&= \frac{1}{Z_B} \text{Tr}_B \left(e^{-\beta \sum_j \hbar \omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right)} e^{-\sum_j \frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar \omega_j} (\hat{b}_j^\dagger - \hat{b}_j)} \right) \\
&= \prod_j \frac{1}{Z_{B,j}} \text{Tr} \left(e^{-\beta \hbar \omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right)} e^{-\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar \omega_j} (\hat{b}_j^\dagger - \hat{b}_j)} \right)
\end{aligned} \tag{32}$$

In the last equality we have used the fact that the creation and annihilation operators of different bath modes commute. $Z_B = \prod_j Z_{B,j}$ is the partition function of the bath and $Z_{B,j}$ is the partition

function of mode j . Let $|M_j\rangle$ be the eigenstate of the Hamiltonian of mode j , the trace can be calculated as

$$\begin{aligned}
& \frac{1}{Z_{B,j}} \text{Tr} \left(e^{\hbar\omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right)} e^{-\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j)} \right) \\
&= \sum_{M_j} f(E_{M_j}) \langle M_j | e^{-\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} (\hat{b}_j^\dagger - \hat{b}_j)} | M_j \rangle \\
&= e^{-\frac{1}{2} \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^2} \sum_{M_j} f(E_{M_j}) \langle M_j | e^{-\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \hat{b}_j^\dagger} e^{\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \hat{b}_j} | M_j \rangle \\
&= e^{-\frac{1}{2} \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^2} \sum_{M_j, N_j} f(E_{M_j}) \langle M_j | e^{-\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \hat{b}_j^\dagger} | N_j \rangle \langle N_j | e^{\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \hat{b}_j} | M_j \rangle
\end{aligned} \tag{33}$$

where $f(E_{M_j})$ is the Boltzmann distribution of the energy of mode j . In the second equality, We have used the Baker–Campbell–Hausdorff formula,

$$e^{\hat{X}} e^{\hat{Y}} = e^{\hat{X} + \hat{Y} + \frac{1}{2} [\hat{X}, \hat{Y}] + \frac{1}{12} [\hat{X}, [\hat{X}, \hat{Y}]] - \frac{1}{12} [\hat{Y}, [\hat{X}, \hat{Y}]] + \dots} \tag{34}$$

and $[\hat{b}, [\hat{b}, \hat{b}^\dagger]] = [\hat{b}^\dagger, [\hat{b}, \hat{b}^\dagger]] = 0$.

$$\begin{aligned}
\langle N_j | e^{\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \hat{b}_j} | M_j \rangle &= \sum_l \frac{1}{l!} \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^l \langle N_j | \left(\hat{b}_j \right)^l | M_j \rangle \\
&= \sum_l \frac{1}{l!} \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^l \sqrt{\frac{M_j!}{(M_j - l)!}} \delta_{N_j, M_j - l}
\end{aligned} \tag{35}$$

In the last line, we changed the upper limit of the summation because $(\hat{b}_j)^l | M_j \rangle = 0$ if $M_j < l$. Similarly,

$$\langle M_j | e^{-\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \hat{b}_j^\dagger} | N_j \rangle = \sum_{l'} \frac{1}{l'!} \left(-\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^{l'} \sqrt{\frac{M_j!}{(M_j - l')!}} \delta_{N_j, M_j - l'} \tag{36}$$

Thus

$$\begin{aligned}
& \sum_{N_j} \langle M_j | e^{-\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \hat{b}_j^\dagger} | N_j \rangle \langle N_j | e^{\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \hat{b}_j} | M_j \rangle \\
&= \sum_{N_j} \sum_{l, l'} \frac{(-1)^l}{l! l'!} \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^{l+l'} \sqrt{\frac{M_j!}{(M_j - l)!}} \sqrt{\frac{M_j!}{(M_j - l')!}} \delta_{N_j, M_j - l} \delta_{N_j, M_j - l'} \\
&= \sum_{N_j} \sum_{l, l'} \frac{(-1)^l}{l! l'!} \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^{l+l'} \sqrt{\frac{M_j!}{(M_j - l)!}} \sqrt{\frac{M_j!}{(M_j - l')!}} \delta_{N_j, M_j - l} \delta_{l, l'} \\
&= \sum_l \frac{(-1)^l}{l!} \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^{2l} \frac{M_j!}{l! (M_j - l)!}
\end{aligned} \tag{37}$$

In the second line, the product $\delta_{N_j, M_j - l} \delta_{N_j, M_j - l'}$ is non-zero only when $N_j = M_j - l = M_j - l'$, indicating $l = l'$.

$$\text{Eq. (33)} = e^{-\frac{1}{2} \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^2} \sum_{M_j} f(E_{M_j}) \sum_l \frac{(-1)^l}{l!} \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^{2l} \frac{M_j!}{l!(M_j - l)!} \quad (38)$$

The Laguerre polynomial of order M is defined as

$$L_M(x) = \sum_l^M \frac{(-1)^l x^l}{l!} \frac{M!}{l!(M - l)!} \quad (39)$$

It has the property

$$\sum_M t^M L_M(x) = \frac{1}{1 - t} e^{-\frac{tx}{1-t}} \quad (40)$$

Also note that

$$f(E_{M_j}) = (1 - e^{-\beta\hbar\omega_j}) e^{-M_j\beta\hbar\omega_j} = (1 - e^{-\beta\hbar\omega_j}) \left(e^{-\beta\hbar\omega_j} \right)^{M_j} \quad (41)$$

We then have

$$\begin{aligned} \text{Eq. (33)} &= \exp \left[-\frac{1}{2} \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^2 - \frac{e^{-\beta\hbar\omega_j}}{1 - e^{-\beta\hbar\omega_j}} \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^2 \right] \\ &= \exp \left[-\frac{1}{2} \coth \left(\frac{\beta\hbar\omega_j}{2} \right) \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^2 \right] \end{aligned} \quad (42)$$

Substituting this expression back into Eq. (32), we obtain the expression of κ_{mn} :

$$\kappa_{mn} = \exp \left[-\frac{1}{2} \sum_j \coth \left(\frac{\beta\hbar\omega_j}{2} \right) \left(\frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar\omega_j} \right)^2 \right] \quad (43)$$

or in terms of the spectral density,

$$\kappa_{mn} = \exp \left[-\frac{1}{2\pi\hbar} \int_0^\infty d\omega \coth \left(\frac{\beta\hbar\omega}{2} \right) \frac{J_{mm}(\omega) + J_{nn}(\omega) - 2J_{mn}(\omega)}{\omega^2} \right] \quad (44)$$

The polaron transformed Hamiltonian can be re-partitioned as

$$\hat{\tilde{H}} = \hat{\tilde{H}}_S + \hat{\tilde{H}}_B + \hat{\tilde{H}}_{SB} \quad (45)$$

where

$$\begin{aligned} \hat{\tilde{H}}_S &= \sum_m \varepsilon_m |m\rangle \langle m| + \sum_{m \neq n} \kappa_{mn} V_{mn} |m\rangle \langle n| \\ \hat{\tilde{H}}_B &= \sum_j \hbar\omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right) \\ \hat{\tilde{H}}_{SB} &= \sum_{m \neq n} V_{mn} \left(\hat{\theta}_m^\dagger \hat{\theta}_n - \kappa_{mn} \right) |m\rangle \langle n| \end{aligned} \quad (46)$$

3 Bath Correlation Function

After polaron transformation, the system-bath coupling can be still factorized as $\hat{\tilde{H}}_{\text{SB}} = \sum_{m,n} \hat{\tilde{S}}_{mn} \otimes \hat{\tilde{B}}_{mn}$, where

$$\hat{\tilde{S}}_{mn} = V_{mn} |m\rangle \langle n| \quad (47)$$

$$\hat{\tilde{B}}_{mn} = \hat{\theta}_m^\dagger \hat{\theta}_n - \kappa_{mn} \quad (48)$$

The bath correlation function is defined as

$$\tilde{C}_{mn,m'n'}(t) = \text{Tr}_{\text{B}} \left(\hat{\tilde{B}}_{mn}(t) \hat{\tilde{B}}_{m'n'}(0) \hat{\rho}_{\text{B,eq}} \right) \quad (49)$$

where $\hat{\tilde{B}}_{mn}(t) = e^{\frac{i}{\hbar} \hat{\tilde{H}}_{\text{B}} t} \hat{\tilde{B}}_{mn} e^{-\frac{i}{\hbar} \hat{\tilde{H}}_{\text{B}} t}$, and $\hat{\rho}_{\text{B,eq}} = \frac{e^{-\beta \hat{\tilde{H}}_{\text{B}}}}{\text{Tr}_{\text{B}} e^{-\beta \hat{\tilde{H}}_{\text{B}}}}$.

$\tilde{C}_{mn,m'n'}(t)$ can be calculated analytically following the same procedure as the calculation of κ_{mn} . For notational simplicity, we define $g_{j,mn} = \frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar \omega_j}$.

$$\begin{aligned} \tilde{C}_{mn,m'n'}(t) &= \text{Tr}_{\text{B}} \left[e^{\frac{i}{\hbar} \hat{\tilde{H}}_{\text{B}} t} \left(\hat{\theta}_m^\dagger \hat{\theta}_n - \kappa_{mn} \right) e^{-\frac{i}{\hbar} \hat{\tilde{H}}_{\text{B}} t} \left(\hat{\theta}_{m'}^\dagger \hat{\theta}_{n'} - \kappa_{m'n'} \right) \hat{\rho}_{\text{B,eq}} \right] \\ &= \text{Tr}_{\text{B}} \left[e^{\frac{i}{\hbar} \hat{\tilde{H}}_{\text{B}} t} \left(\hat{\theta}_m^\dagger \hat{\theta}_n \right) e^{-\frac{i}{\hbar} \hat{\tilde{H}}_{\text{B}} t} \left(\hat{\theta}_{m'}^\dagger \hat{\theta}_{n'} \right) \hat{\rho}_{\text{B,eq}} \right] - \kappa_{mn} \kappa_{m'n'} \end{aligned} \quad (50)$$

The first term

$$\begin{aligned} \text{Tr}_{\text{B}}(\dots) &= \frac{1}{Z_{\text{B}}} \text{Tr}_{\text{B}} \left(e^{-\beta \sum_j \hbar \omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right)} e^{it \sum_j \omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right)} e^{-\sum_j g_{j,mn} (\hat{b}_j^\dagger - \hat{b}_j)} e^{-it \sum_j \omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right)} \right. \\ &\quad \left. \times e^{-\sum_j g_{j,m'n'} (\hat{b}_j^\dagger - \hat{b}_j)} \right) \\ &= \prod_j \frac{1}{Z_{\text{B},j}} \text{Tr} \left(e^{-\beta \hbar \omega_j \left(\hat{b}_j^\dagger \hat{b}_j + \frac{1}{2} \right)} e^{i \omega_j t \hat{b}_j^\dagger \hat{b}_j} e^{-g_{j,mn} (\hat{b}_j^\dagger - \hat{b}_j)} e^{-i \omega_j t \hat{b}_j^\dagger \hat{b}_j} e^{-g_{j,m'n'} (\hat{b}_j^\dagger - \hat{b}_j)} \right) \\ &= \prod_j \sum_{M_j} f(E_{M_j}) \langle M_j | e^{i \omega_j t \hat{b}_j^\dagger \hat{b}_j} e^{-g_{j,mn} (\hat{b}_j^\dagger - \hat{b}_j)} e^{-i \omega_j t \hat{b}_j^\dagger \hat{b}_j} e^{-g_{j,m'n'} (\hat{b}_j^\dagger - \hat{b}_j)} | M_j \rangle \end{aligned} \quad (51)$$

Note that for any unitary operator $\hat{U}$ and any operator $\hat{O}$, the following identity holds,

$$\hat{U}^\dagger e^{\hat{O}} \hat{U} = e^{\hat{U}^\dagger \hat{O} \hat{U}} \quad (52)$$

Let

$$\hat{b}_j(t) \equiv e^{i \omega_j t \hat{b}_j^\dagger \hat{b}_j} \hat{b}_j e^{-i \omega_j t \hat{b}_j^\dagger \hat{b}_j}, \quad \hat{b}_j^\dagger(t) \equiv e^{i \omega_j t \hat{b}_j^\dagger \hat{b}_j} \hat{b}_j^\dagger e^{-i \omega_j t \hat{b}_j^\dagger \hat{b}_j} \quad (53)$$

Applying $\hat{b}_j(t)$ and $\hat{b}_j^\dagger(t)$ to an arbitrary state, which can be written as linear combination of the eigenstates $|M_j\rangle$, after a few lines of calculation one can show that

$$\hat{b}_j(t) = e^{-i \omega_j t} \hat{b}_j, \quad \hat{b}_j^\dagger(t) = e^{i \omega_j t} \hat{b}_j^\dagger \quad (54)$$

We further define

$$g_{j,mn}(t) = g_{j,mn} e^{-i \omega_j t} \quad (55)$$

Then

$$\begin{aligned} & \langle M_j | e^{i\omega_j t \hat{b}_j^\dagger \hat{b}_j} e^{-g_{j,mn}(\hat{b}_j^\dagger - \hat{b}_j)} e^{-i\omega_j t \hat{b}_j^\dagger \hat{b}_j} e^{-g_{j,m'n'}(\hat{b}_j^\dagger - \hat{b}_j)} | M_j \rangle \\ &= \langle M_j | e^{-g_{j,mn}^*(t) \hat{b}_j^\dagger + g_{j,mn}(t) \hat{b}_j} e^{-g_{j,m'n'}^*(t) \hat{b}_j^\dagger + g_{j,m'n'}(0) \hat{b}_j} | M_j \rangle \end{aligned} \quad (56)$$

Again, using the Baker–Campbell–Hausdorff formula and the commutation relation,

$$e^{-g_{j,mn}^*(t) \hat{b}_j^\dagger + g_{j,mn}(t) \hat{b}_j} = e^{-\frac{1}{2} g_{j,mn}^2} e^{-g_{j,mn}^*(t) \hat{b}_j^\dagger} e^{g_{j,mn}(t) \hat{b}_j} \quad (57)$$

$$e^{-g_{j,m'n'}^*(0) \hat{b}_j^\dagger + g_{j,m'n'}(0) \hat{b}_j} = e^{-\frac{1}{2} g_{j,m'n'}^2} e^{-g_{j,m'n'}^*(0) \hat{b}_j^\dagger} e^{g_{j,m'n'}(0) \hat{b}_j} \quad (58)$$

$$e^{g_{j,mn}(t) \hat{b}_j} e^{-g_{j,m'n'}^*(0) \hat{b}_j^\dagger} = e^{-g_{j,mn}(t) g_{j,m'n'}^*(0)} e^{-g_{j,m'n'}^*(0) \hat{b}_j^\dagger} e^{g_{j,mn}(t) \hat{b}_j} \quad (59)$$

Thus

$$\begin{aligned} & \langle M_j | e^{-g_{j,mn}^*(t) \hat{b}_j^\dagger + g_{j,mn}(t) \hat{b}_j} e^{-g_{j,m'n'}^*(t) \hat{b}_j^\dagger + g_{j,m'n'}(0) \hat{b}_j} | M_j \rangle \\ &= e^{-\frac{1}{2} [g_{j,mn}^2 + g_{j,m'n'}^2 + 2g_{j,mn}(t) g_{j,m'n'}^*(0)]} \langle M_j | e^{-[g_{j,mn}^*(t) + g_{j,m'n'}^*(0)] \hat{b}_j^\dagger} e^{[g_{j,mn}(t) + g_{j,m'n'}(0)] \hat{b}_j} | M_j \rangle \\ &= e^{-\frac{1}{2} [g_{j,mn}^2 + g_{j,m'n'}^2 + 2g_{j,mn}(t) g_{j,m'n'}^*(0)]} \langle M_j | e^{-\xi^*(t) \hat{b}_j^\dagger} e^{\xi(t) \hat{b}_j} | M_j \rangle \\ &= e^{-\frac{1}{2} [g_{j,mn}^2 + g_{j,m'n'}^2 + 2g_{j,mn}(t) g_{j,m'n'}^*(0)]} L_{M_j}(|\xi(t)|^2) \end{aligned} \quad (60)$$

where $\xi(t) \equiv g_{j,mn}(t) + g_{j,m'n'}(0)$. The thermal average can be calculated using Eq. (40) as

$$\begin{aligned} & \sum_{M_j} f(E_{M_j}) \langle M_j | e^{i\omega_j t \hat{b}_j^\dagger \hat{b}_j} e^{-g_{j,mn}(\hat{b}_j^\dagger - \hat{b}_j)} e^{-i\omega_j t \hat{b}_j^\dagger \hat{b}_j} e^{-g_{j,m'n'}(\hat{b}_j^\dagger - \hat{b}_j)} | M_j \rangle \\ &= e^{-\frac{1}{2} [g_{j,mn}^2 + g_{j,m'n'}^2 + 2g_{j,mn}(t) g_{j,m'n'}^*(0)]} e^{-\frac{e^{-\beta \hbar \omega_j}}{1 - e^{-\beta \hbar \omega_j}} |g_{j,mn}(t) + g_{j,m'n'}(0)|^2} \\ &= e^{-\frac{1}{2} [g_{j,mn}^2 + g_{j,m'n'}^2 + 2g_{j,mn}(t) g_{j,m'n'}^*(0)]} e^{-\frac{e^{-\beta \hbar \omega_j}}{1 - e^{-\beta \hbar \omega_j}} [g_{j,mn}^2 + g_{j,m'n'}^2 + g_{j,mn}(t)^* g_{j,m'n'}(0) + g_{j,mn}(t) g_{j,m'n'}(0)^*]} \\ &= e^{-\frac{1}{2} \coth\left(\frac{\beta \hbar \omega_j}{2}\right) g_{j,mn}^2} e^{-\frac{1}{2} \coth\left(\frac{\beta \hbar \omega_j}{2}\right) g_{j,m'n'}^2} \\ & \quad \times \exp \left\{ g_{j,mn} g_{j,m'n'} \left[-e^{-i\omega_j t} - \frac{e^{-\beta \hbar \omega_j}}{1 - e^{-\beta \hbar \omega_j}} (e^{i\omega_j t} + e^{-i\omega_j t}) \right] \right\} \\ &= e^{-\frac{1}{2} \coth\left(\frac{\beta \hbar \omega_j}{2}\right) g_{j,mn}^2} e^{-\frac{1}{2} \coth\left(\frac{\beta \hbar \omega_j}{2}\right) g_{j,m'n'}^2} \\ & \quad \times \exp \left\{ -g_{j,mn} g_{j,m'n'} \left[\cos(\omega_j t) \coth\left(\frac{\beta \hbar \omega_j}{2}\right) - i \sin(\omega_j t) \right] \right\} \end{aligned} \quad (61)$$

Performing the production with respect to j , we get the final expression of $\tilde{C}_{mn,m'n'}(t)$,

$$\tilde{C}_{mn,m'n'}(t) = \kappa_{mn} \kappa_{m'n'} \left[e^{-\sum_j \frac{\gamma_{j,m} - \gamma_{j,n}}{\hbar \omega_j} \frac{\gamma_{j,m'} - \gamma_{j,n'}}{\hbar \omega_j} \left[\cos(\omega_j t) \coth\left(\frac{\beta \hbar \omega_j}{2}\right) - i \sin(\omega_j t) \right]} - 1 \right] \quad (62)$$

or in terms of the spectral density,

$$\tilde{C}_{mn,m'n'}(t) = \kappa_{mn} \kappa_{m'n'} \left[e^{-\frac{1}{\pi \hbar} \int_0^\infty d\omega \frac{J_{mm'}(\omega) + J_{nn'}(\omega) - J_{mn'}(\omega) - J_{nm'}(\omega)}{\omega^2} [\cos(\omega t) \coth\left(\frac{\beta \hbar \omega}{2}\right) - i \sin(\omega t)]} - 1 \right] \quad (63)$$

4 Polaron Transformed Redfield Equation

Let $\hat{\rho}_{\text{tot}}$ be the total density operator of the system and the bath in the original frame, the reduced density operator of the system in polaron frame is

$$\hat{\tilde{\rho}} = \text{Tr}_B \left(\hat{\tilde{\rho}}_{\text{tot}} \right) = \text{Tr}_B \left(e^{\hat{G}} \hat{\rho}_{\text{tot}} e^{-\hat{G}} \right) \quad (64)$$

Note that $\hat{\tilde{\rho}} \neq e^{\hat{G}} \hat{\rho} e^{-\hat{G}}$ because polaron transformation mixes system and bath degree of freedoms. In general, this transformation leads to some complicated entanglement between the system and the bath. But **the population of the system is invariant**. The population of state $|m\rangle$ can be calculated as

$$\begin{aligned} \rho_{mm} &= \text{Tr}_{S+B} (|m\rangle\langle m| \hat{\rho}_{\text{tot}}) \\ &= \text{Tr}_{S+B} \left(e^{-\hat{G}} e^{\hat{G}} |m\rangle\langle m| e^{-\hat{G}} e^{\hat{G}} \hat{\rho}_{\text{tot}} \right) \\ &= \text{Tr}_{S+B} \left(|m\rangle\langle m| e^{\hat{G}} \hat{\rho}_{\text{tot}} e^{-\hat{G}} \right) \\ &= \text{Tr}_{S+B} \left(|m\rangle\langle m| \hat{\tilde{\rho}}_{\text{tot}} \right) = \tilde{\rho}_{mm} \end{aligned} \quad (65)$$

The derivation of quantum master equation and Redfield equation in the polaron frame follows the same procedure as in the original frame. In the derivation, we made assumption that there is no initial correlation and the initial density operator of the total system can be factorized as

$$\hat{\tilde{\rho}}_{\text{tot}}(0) = \hat{\tilde{\rho}}(0) \otimes \hat{\tilde{\rho}}_{B,\text{eq}} \quad (66)$$

This is generally not valid in the polaron frame because the system and bath become entangled after polaron transformation. Regardless, in most cases, the decoupled initial condition is only an approximation and only impact the short-time dynamics.

Let $|a\rangle, |b\rangle, \dots$ be the eigenstate of the polaron transformed system Hamiltonian $\hat{\tilde{H}}_S$, $E_a, E_b, \dots$ be the energies of the eigenstates, and $\omega_{ab} = (E_a - E_b)/\hbar$. The polaron transformed Redfield equation take the same form as in the original frame

$$\frac{d}{dt} \tilde{\rho}_{ab} = -i\omega_{ab} \tilde{\rho}_{ab} + \frac{1}{\hbar^2} \sum_{c,d} \mathcal{R}_{abcd} \tilde{\rho}_{cd} \quad (67)$$

The polaron transformed Redfield tensor also takes the same form

$$\boxed{\mathcal{R}_{abcd} = -\delta_{bd} \sum_e \Gamma_{aeec} - \delta_{ac} \sum_e \Gamma_{beed}^* + \Gamma_{dbac} + \Gamma_{cabd}^*} \quad (68)$$

The difference is in the expression of the Γ tensors. In the polaron frame,

$$\boxed{\Gamma_{abcd} = \sum_{mn,m'n'} \tilde{S}_{ab}^{(mn)} \tilde{S}_{cd}^{(m'n')} \tilde{G}_{mn,m'n'}(\omega_{dc})} \quad (69)$$

where

$$\boxed{\tilde{S}_{ab}^{(mn)} = \langle a | \tilde{S}_{mn} | b \rangle = V_{mn} \langle a | m \rangle \langle n | b \rangle} \quad (70)$$

$$\boxed{\tilde{G}_{mn,m'n'}(\omega) = \int_0^\infty dt e^{i\omega t} \tilde{C}_{mn,m'n'}(t)} \quad (71)$$